

BELLINGSHAUSEN STATION, AntarcticaWMO: **890500**Lat: **62.192S**Lon: **58.936W**Elev: **14**StdP: **101.16**Time Zone: **-3.00 (W03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-18.4	-15.9	-20.8	0.6	-18.1	-18.3	0.7	-15.6	38.2	-12.3	33.8	-11.4	10.6	160	1.132

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.0	4.2	3.4	3.7	2.9	3.3	2.6	3.7	4.0	3.2	3.5	2.8	3.1	8.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
3.3	4.8	3.7	2.9	4.6	3.3	2.5	4.5	3.0	16.4	4.2	15.5	3.6	14.8	3.2	6.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	21.8	18.3	DB	-19.2	5.8	3.3	1.3	-21.6	6.7	-23.5	7.4	-25.4	8.1	-27.8	9.0	(4)
(5)			WB	-19.7	4.7	3.4	0.8	-22.1	5.3	-24.1	5.7	-26.0	6.1	-28.5	6.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.2	1.3	1.5	0.6	-1.7	-2.6	-4.8	-6.1	-5.9	-4.6	-2.5	-1.3	0.2
	DBStd	3.96	1.13	1.44	1.88	3.02	2.85	3.90	4.67	4.48	3.83	2.30	1.66	1.24
	HDD10.0	4444	269	238	291	351	392	445	498	493	437	387	338	304
	HDD18.3	7486	528	472	549	601	650	695	757	751	687	646	588	563
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.9	6.9	7.1	7.4	8.4	7.9	8.5	8.3	8.4	8.4	8.7	7.7	6.8
Precipitation	PrecAvg	1040	82	81	120	102	96	77	72	75	82	98	84	70
	PrecMax	1392	115	125	187	153	161	114	109	142	137	171	115	114
	PrecMin	856	48	49	71	58	42	40	33	15	29	51	31	31
	PrecStd	117	14	19	29	23	28	19	23	27	24	26	20	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	5.1	5.0	3.5	2.6	1.7	1.0	0.9	1.2	1.6	2.7	3.9
		MCWB	4.1	4.2	4.1	2.8	1.8	1.0	0.4	0.2	0.5	0.9	1.8	3.0
	2%	DB	3.9	4.2	3.9	2.7	1.9	1.0	0.4	0.2	0.6	0.9	1.7	2.9
		MCWB	3.0	3.5	3.2	2.1	1.3	0.5	-0.1	-0.2	0.1	0.3	1.0	2.0
	5%	DB	3.4	3.7	3.4	2.1	1.4	0.5	-0.1	-0.2	0.1	0.5	1.1	2.3
		MCWB	2.6	3.1	2.8	1.6	0.9	0.1	-0.5	-0.5	-0.2	0.0	0.6	1.5
	10%	DB	2.9	3.3	3.0	1.6	0.9	-0.1	-0.6	-0.6	-0.2	0.1	0.7	1.9
		MCWB	2.3	2.7	2.5	1.1	0.5	-0.4	-1.0	-0.9	-0.5	-0.3	0.2	1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	4.5	4.2	3.0	2.1	1.2	0.6	0.5	0.9	1.1	2.1	3.2
		MCDB	4.7	4.9	4.6	3.3	2.3	1.3	0.7	0.7	1.1	1.3	2.5	3.6
	2%	WB	3.4	3.7	3.5	2.2	1.5	0.6	0.1	0.0	0.2	0.6	1.2	2.3
		MCDB	3.7	4.0	3.9	2.5	1.7	0.8	0.3	0.1	0.4	0.7	1.5	2.6
	5%	WB	2.9	3.2	3.0	1.7	1.1	0.2	-0.4	-0.4	-0.1	0.2	0.7	1.8
		MCDB	3.1	3.5	3.3	2.0	1.3	0.4	-0.2	-0.3	0.1	0.3	0.9	2.1
	10%	WB	2.5	2.9	2.6	1.3	0.6	-0.3	-0.9	-0.9	-0.4	-0.2	0.4	1.3
		MCDB	2.8	3.2	2.9	1.5	0.8	-0.1	-0.7	-0.7	-0.3	0.0	0.6	1.7
Mean Daily Temperature Range	5% DB	MDBR	1.9	2.0	2.3	2.7	3.0	3.6	4.4	4.0	3.7	2.5	2.0	1.8
		MCDBR	2.5	2.4	2.4	2.4	2.8	3.0	3.4	3.2	2.8	2.1	2.1	2.3
		MCWBR	2.1	2.0	2.3	2.4	2.6	3.0	3.4	3.2	2.9	2.1	2.0	2.0
	5% WB	MCDBR	2.3	2.2	2.3	2.3	2.6	2.8	3.4	3.2	2.7	2.0	2.0	2.2
		MCWBR	2.2	2.1	2.3	2.4	2.6	2.9	3.6	3.3	2.9	2.1	2.0	2.0
Clear-Sky Solar Irradiance	taub	0.309	0.308	0.297	0.280	0.225	0.366	0.211	0.269	0.288	0.310	0.304	0.309	
	taud	2.493	2.501	2.501	2.373	2.100	2.821	2.049	2.239	2.369	2.400	2.457	2.465	
	Ebn at Noon	967	918	842	676	443	302	438	660	828	903	966	978	
	Edh at Noon	98	87	71	53	31	20	31	57	78	96	101	104	
All-Sky Solar Radiation	RadAvg	5.01	3.61	1.95	0.85	0.25	0.10	0.18	0.63	1.67	3.09	4.74	5.67	
	RadStd	0.46	0.28	0.16	0.07	0.02	0.00	0.01	0.06	0.13	0.21	0.29	0.45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	-0.26	-0.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page